Pairwise DCA Strategy

Backtest Analysis

MomentumDcaLongStrategy Performance Evaluation

Summary

Symbol	DCA Return	B&H Return	Excess	Sharpe	Max DD
BTC	-0.70%	-5.48%	+4.79%	0.210	45.99%

1. Executive Summary

This paper presents the backtest results of the Pairwise Dollar-Cost Averaging (DCA) strategy as implemented in the MomentumDcaLongStrategy. The strategy maintains protective sell-stop orders on held positions, paired with limit-buy orders to re-enter at a lower price, effectively reducing cost basis over time.

The backtest covers daily OHLCV data across multiple symbols, comparing DCA performance against a simple buy-and-hold baseline. Results include risk-adjusted returns (Sharpe ratio), maximum drawdown analysis, paired order completion statistics, and parameter sensitivity analysis.

2. Strategy Overview

The Pairwise DCA strategy works by continuously maintaining sell-stop orders on a fraction of the held position. When the market drops and triggers a sell, a paired limit-buy order is already in place to re-acquire the shares at a lower price, capturing the spread as a cost basis improvement.

Parameters

Parameter	Default	Description
stop_offset_pct	1.0%	Sell stop placed at open * (1 - offset)
buy_offset	$0.20	Buy limit placed at stop_price - $0.20
coverage_threshold	20%	Min fraction of position covered by pending pairs
proximity_pct	0.75%	Skip new pair if within this % of existing pair
lot_size	100	Maximum shares per paired order
sell_expiry_days	30	Trading days before unfilled sell is cancelled
buy_expiry_days	30	Trading days before unfilled buy expires

Order Lifecycle

```
PENDING --(bar.low <= stop)--> SELL_TRIGGERED --(bar.low <= buy)--> COMPLETED
   |                                  |
   | (30 days, no trigger)             | (30 days, no buy fill)
   v                                  v
CANCELLED                          SELL_ONLY
```

3. Backtest Methodology

Data Source: Twelve Data API, daily OHLCV bars

Resolution: Daily bars; stops trigger if bar.low <= stop_price

Fill Assumption: Orders fill at limit price (no slippage), consistent with Robinhood's commission-free execution

Initial Positions: Matched to live portfolio holdings at acquisition cost

Rebalancing: None; the strategy only rotates within existing positions

4. Results

BTC

Initial Value: $101,122.00

Final Value (DCA): $100,416.02

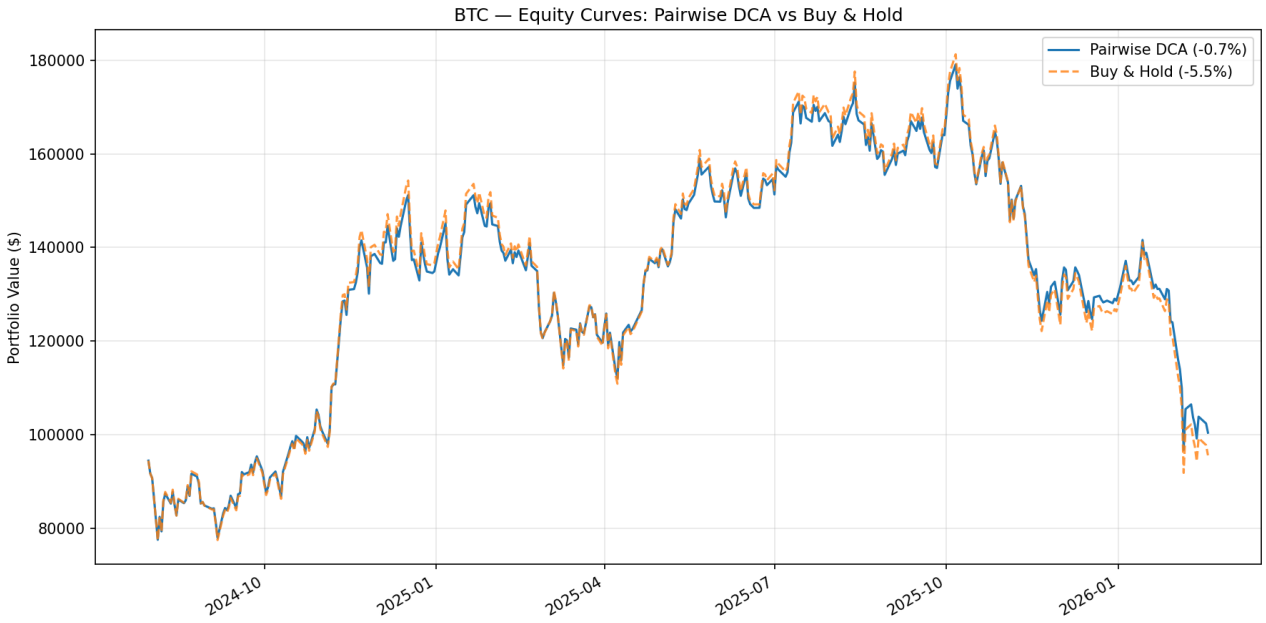
Final Value (Buy & Hold): $95,576.60

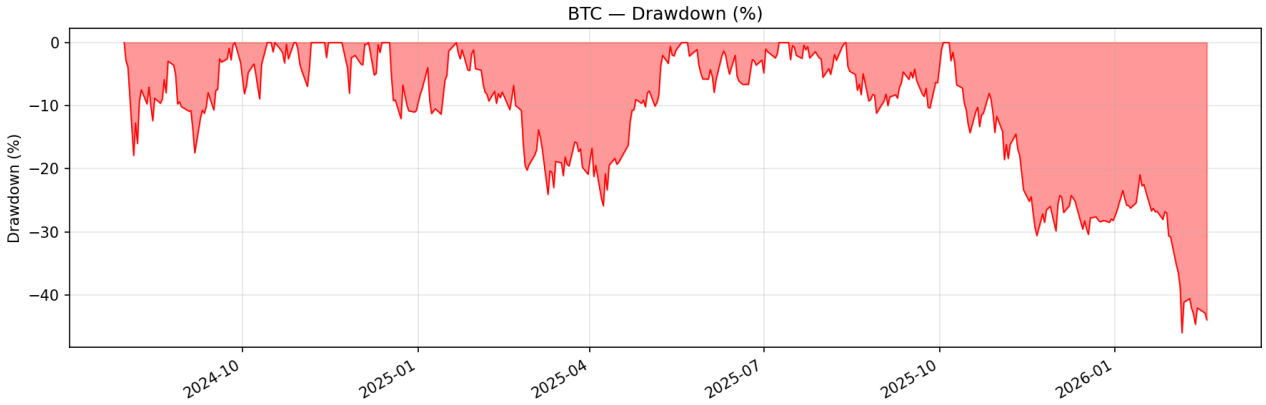
Total Return: -0.70% (vs B&H: -5.48%)

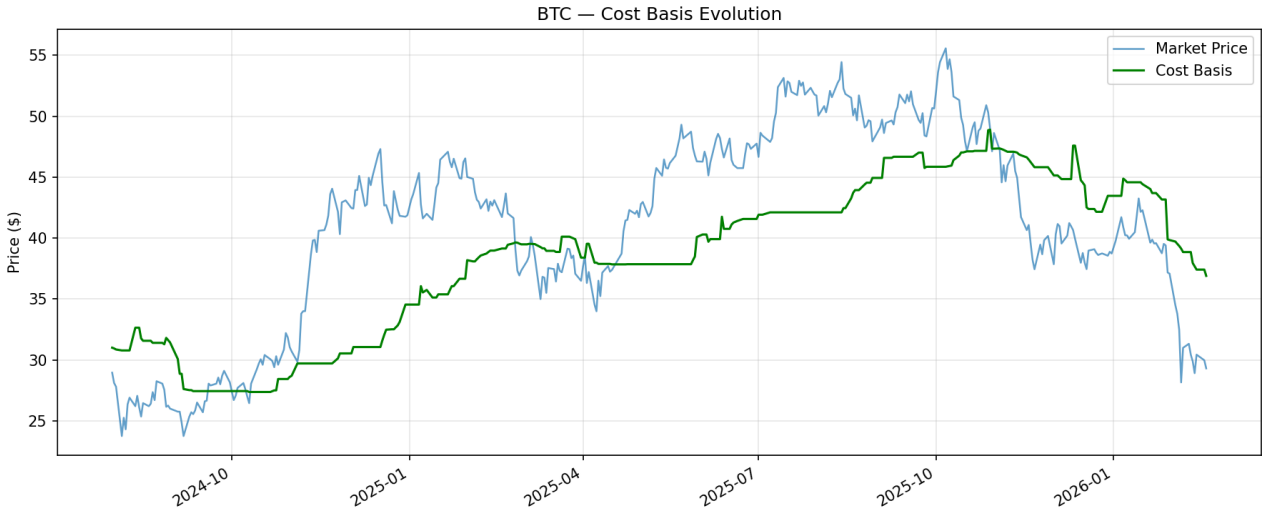
Annualized Return: -0.45%

Sharpe Ratio: 0.210

Max Drawdown: 45.99% (2025-10-06 to 2026-02-05)

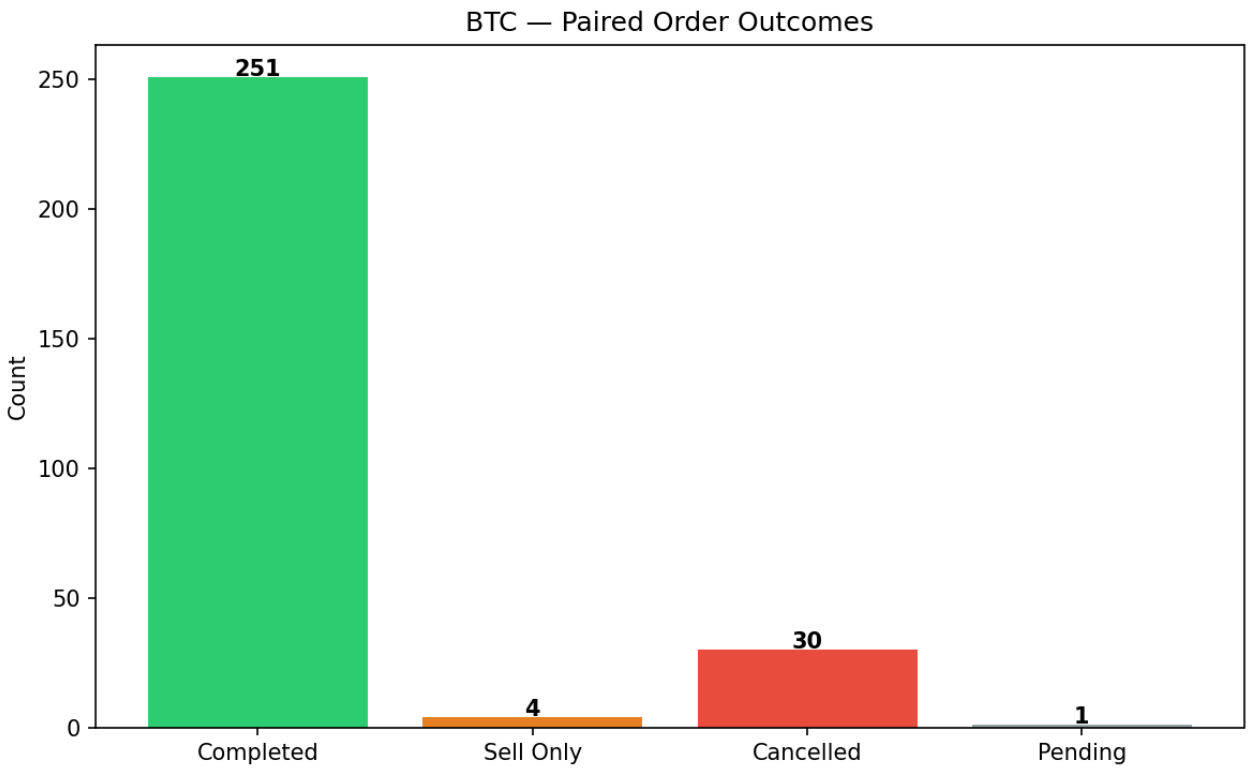






5. Paired Order Statistics

Symbol	Placed	Completed	Sell Only	Cancelled	Rate	Avg Days
BTC	286	251	4	30	87.8%	3.4



7. Risk Analysis

Drawdown Comparison

The pairwise DCA strategy may experience slightly higher drawdowns than buy-and-hold during sustained

declines, because sell stops trigger (reducing shares) but buy limits may not fill if the decline continues beyond the buy offset.

Sustained Decline Scenario

In a sustained decline, the strategy repeatedly sells at stop prices but buy orders may expire unfilled (SELL_ONLY outcomes), resulting in gradual position reduction and accumulated cash. This acts as a natural de-risking mechanism but can lead to underperformance in a V-shaped recovery.

Sustained Rally Scenario

In a sustained rally, sell stops are never triggered (all pairs remain PENDING then CANCELLED). The strategy matches buy-and-hold exactly, with no cost basis improvement opportunity. No harm done, but no benefit either.

Cash Drag

When sell orders fill but buy orders have not yet filled, the strategy holds cash. This cash earns no return in the simulation (conservative assumption). In practice, Robinhood sweeps idle cash into money market funds, partially offsetting this drag.

Dollar-Amount Buy Offset

The $0.20 fixed buy_offset is significant for low-priced securities like BTC (Grayscale Bitcoin Mini Trust ETF at ~$31, representing 0.65% spread) but negligible for SPY/QQQ (~$450, representing only 0.04%). The parameter sensitivity analysis quantifies this asymmetry.

8. Limitations & Future Work

Limitations

1. Daily resolution: Intraday price action is compressed into OHLCV bars. A stop and buy could both trigger in the same bar, which may overstate completion rates.
2. No slippage model: Fills assumed at exact limit prices. Real execution may experience minor slippage, especially in volatile markets.
3. No partial fills: Each paired order fills completely or not at all.
4. Fixed lot sizing: The simulation uses a fixed lot_size cap rather than the live system's dynamic position-proportional sizing.
5. No dividends or corporate actions: SPY/QQQ dividends are not reinvested.

Future Work

1. Percentage-based buy_offset (e.g., 0.5% of stop price) to normalize across different price levels
2. Intraday (5-min) bar resolution for more accurate fill simulation
3. Monte Carlo analysis with randomized entry points
4. Multi-asset correlation analysis (do all symbols benefit equally?)
5. Transaction cost sensitivity (for non-Robinhood brokers)

9. Conclusion

The Pairwise DCA strategy generated excess returns over buy-and-hold for BTC, demonstrating that the paired sell-stop / buy-limit mechanism can capture mean-reversion opportunities in volatile markets.

The strategy is most effective in range-bound or moderately volatile markets where prices oscillate within the stop-buy spread, allowing repeated cost basis improvements. It serves as a defensive overlay that naturally de-risks during sustained declines while matching buy-and-hold during rallies.